LESSON 2.9

Describe Data by Mean Absolute Deviation

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the mean absolute deviation (MAD) to describe how spread out data is.

WHY THIS MATTERS The forecast shows the expected high temperatures for a week. The mean high temperature for the week is 79°F. How can you describe the variation in high temperatures over the week?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Mean Absolute Deviation *Spanish: Desviación media absoluta*	The average distance of data values from the mean; it describes spread.	Written as MAD
Deviation *Spanish: Desviación*	How far a number is from the mean.	Mean 19, value 23 → 4
Absolute Value *Spanish: Valor absoluto*	How far a number is from zero. It is never negative.	Distance is never below zero
Spread *Spanish: Dispersión*	How far apart the numbers are.	Small MAD = tight set
Data distribution *Spanish: Distribución de datos*	How the data looks: where it sits and how spread out it is.	A set clustered tightly around the mean has low MAD; a set spread far from the mean has high MAD
Variability *Spanish: Variabilidad*	How spread out the numbers are.	Low variability (MAD = 1): very consistent. High variability (MAD = 8): very spread out

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Four plants grew 3, 7, 8 and 10 cm. Find the MAD.

1. Mean: 28 ÷ 4 = 7.
2. Distances from 7: 4, 0, 1, 3.
3. Sum = 8, and 8 ÷ 4 = 2.

Answer: MAD = 2 cm

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

New data set: 9, 10, 11. First, what is the mean? Add the three numbers and divide by 3.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The mean of this set is _____ .
- The distance from _____ to the mean is _____ .
- The distances total _____ , and there are _____ values.
- The MAD is _____ , which means _____ .

Word bank mean absolute deviation | deviation | distance | absolute value | mean | spread | average | total | divide | typical | consistent | variation

WATCH OUT!

A common mistake

A common mistake in Mean Absolute Deviation is adding the signed deviations (value − mean) instead of their absolute values, so the positives and negatives cancel out to 0. For 6, 8, 10, 12 (mean = 9), the deviations are −3, −1, 1, 3

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

The mean of a data set is 15. One value is 11. What is the absolute deviation of that value from the mean?

(A) 4
(B) −4
(C) 11
(D) 26

STEPS TO TRY

1. Absolute deviation asks: how FAR is the value 11 from the mean of 15? A distance is never negative; zero is allowed.
2. Subtract: value − mean gives the deviation. Then take the absolute value (drop any negative sign) to get the absolute deviation.
3. Step 1: set up 11 − 15 (the result will be negative). Step 2 is turning that result into a positive distance.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A data set has absolute deviations of 3, 1, 5, 2, 4. What is the MAD?

(A) 1
(B) 3
(C) 5
(D) 15

STEPS TO TRY

1. You're given the five absolute deviations already — no subtracting needed. MAD is their average.
2. MAD = (sum of the absolute deviations) ÷ (how many there are). Count them: there are 5 deviations listed.
3. Step 1: add 3 + 1 + 5 + 2 + 4. Step 2: divide that total by 5. Don't stop after adding — the sum alone is not the MAD.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

The mean of a data set is 20. A value is 26. What is the absolute deviation?

(A) 6
(B) −6
(C) 26
(D) 46

STEPS TO TRY

1. Same skill as before: absolute deviation is the distance between the value 26 and the mean 20.
2. Compute value − mean, then take the absolute value. If the difference is already positive, the absolute value keeps it the same.
3. Step 1: set up 26 − 20. Step 2: check the sign — a positive difference is already a distance, so you're nearly done.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Team A has MAD = 2.1 points. Team B has MAD = 6.8 points. Which team is more consistent?

(A) Team B — higher MAD means better performance

(B) Both are equally consistent

(C) Cannot tell from MAD alone

(D) Team A — lower MAD means scores are closer to the mean

STEPS TO TRY

1. Consistent means scores stay close to the team's average game after game. MAD measures exactly that closeness.
2. MAD is the average DISTANCE of scores from the mean. Decide whether a small distance or a large distance means 'more consistent.'
3. Compare what each number says: scores about 2.1 points from the mean vs about 6.8 points away on a typical night. Which team wanders less?

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Player A's MAD is 2.4 points. If Player B's MAD is 7.2 points, what does that tell you about their consistency?

SENTENCE STARTER Player A has a ___ MAD, which means their scores are ___ to the mean. Player B has a ___ MAD, which means their scores are ___ from the mean. The more consistent player is ___ because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help